

ME23312 ENGINEERING THERMODYNAMICS

Unit II – Second Law and Availability Analysis · Presentation 4 of 10

The Carnot Theorem, its Corollaries and the Thermodynamic Temperature Scale

Statement and proof of the Carnot theorem · three corollaries · the Kelvin scale · Carnot relations for any reversible device · using the Carnot limits

In this presentation

1. The Carnot theorem
2. Proof by contradiction
3. Corollary 1 – all reversible engines are equally efficient
4. Corollary 2 – efficiency depends only on the temperatures

1. Corollary 3 – maximum COP
2. The thermodynamic (Kelvin) temperature scale
3. Carnot relations for any reversible device
4. Using the Carnot limits in engineering

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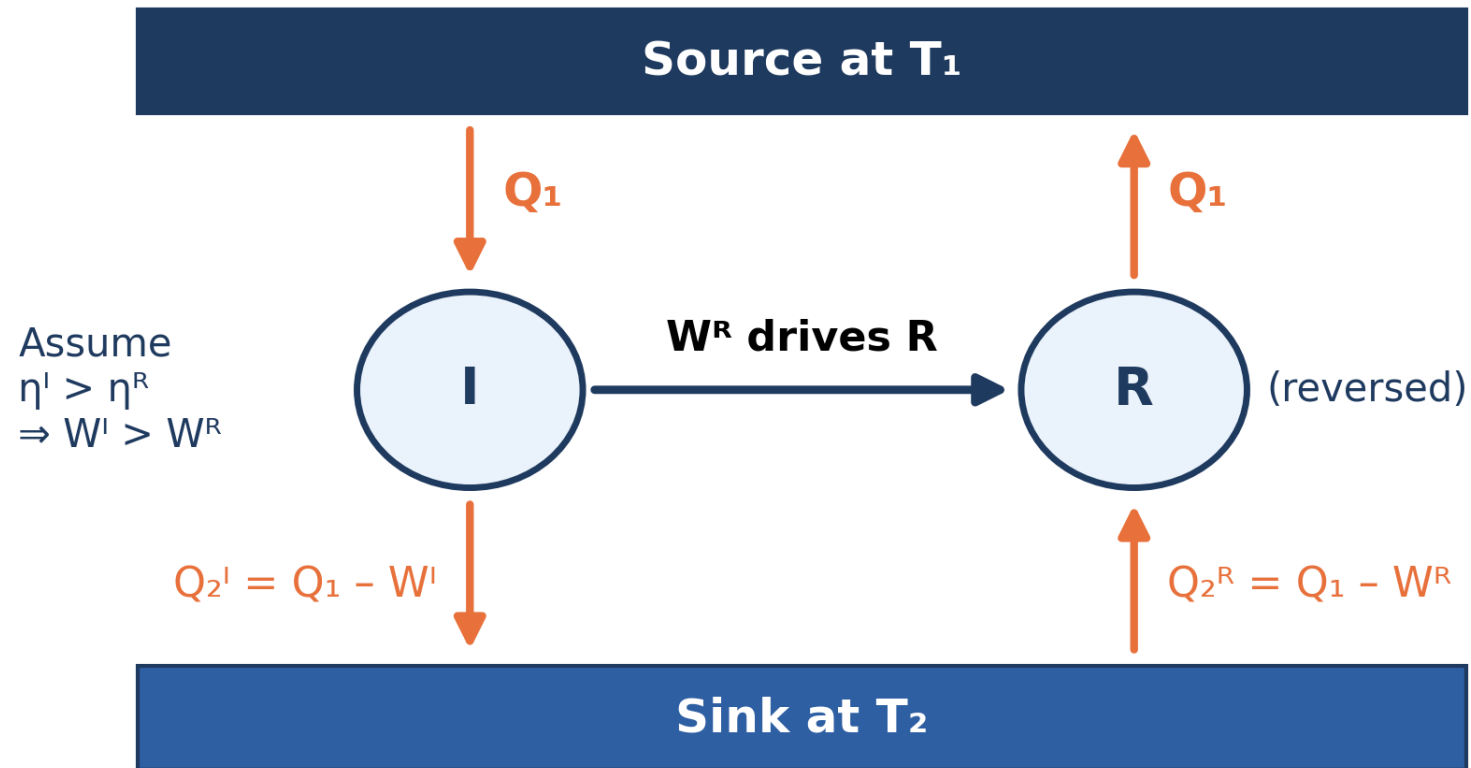
The Carnot theorem

Carnot theorem

No heat engine operating between two given thermal reservoirs can be more efficient than a reversible engine operating between the same two reservoirs.

- $\eta_{\text{irreversible}} \leq \eta_{\text{reversible}}$ for the same source T_1 and sink T_2

Proof of the Carnot theorem



If the irreversible engine I were more efficient than the reversible engine R, part of its work could drive R backwards – the combination would be a PMM2

Proof by contradiction

1. Reversible engine R and irreversible engine I work between the same T_1 and T_2 , each receiving the same Q_1
2. Suppose $\eta_I > \eta_R$: then $W_I > W_R$ and $Q_2^I < Q_2^R$
3. Run R backwards as a refrigerator driven by part W_R of the output of I; R returns Q_1 to the source
4. The source is unaffected – remove it
5. The plant delivers $W_I - W_R > 0$ exchanging heat with the **sink alone** – a PMM2, so $\eta_I \leq \eta_R$

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Corollaries of the Carnot theorem

The three corollaries

1 Same efficiency

All reversible engines between the same two reservoirs have the same efficiency. Apply the theorem to R_A and R_B each way: $\eta_A \leq \eta_B$ and $\eta_B \leq \eta_A$ – independent of working substance and engine details.

2 Temperatures only

The efficiency of a reversible engine depends only on the two reservoir temperatures: $\eta_{\text{rev}} = f(T_1, T_2)$ – the basis of the thermodynamic temperature scale.

3 Maximum COP

No refrigerator or heat pump can have a higher COP than a reversible one between the same reservoirs, and all reversible ones have the same COP. Proof identical in form.

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The thermodynamic temperature scale

Kelvin's argument

- Since η_{rev} depends only on the temperatures, so does the ratio of the heats: $Q_1/Q_2 = F(T_1, T_2)$
- Considering three reservoirs with reversible engines working in pairs, the function must have the form $F = \varphi(T_1)/\varphi(T_2)$
- Kelvin chose the simplest function, $\varphi(T) = T$

The Kelvin (thermodynamic) scale

$$(Q_1/Q_2)_{\text{rev}} = T_1/T_2$$

- The ratio of two temperatures = the ratio of the heats exchanged by a reversible engine with reservoirs at those temperatures
- Independent of the properties of any thermometric substance
- Size of the unit fixed by assigning 273.16 K to the triple point of water

Carnot relations for any reversible device

$$\eta_{\text{rev}} = 1 - T_2/T_1$$

$$(\text{COP})_{\text{ref,rev}} = T_2/(T_1 - T_2) \quad (\text{COP})_{\text{HP,rev}} = T_1/(T_1 - T_2)$$

- Now valid for **any** reversible engine or refrigerator – not only one using an ideal gas
- Absolute zero: the temperature at which a reversible engine would reject no heat

Using the Carnot limits in engineering

- Any real engine: $\eta_{\text{actual}} \leq 1 - T_2/T_1$; any real refrigerator: $\text{COP} \leq T_2/(T_1 - T_2)$
- Test the **feasibility** of a claimed performance
- Find the **least heat rejection** or the **least power** for a given duty
- Rate an actual device against its ideal (second-law efficiency, Presentation 10)

KEY POINT Convert to kelvin first: $T \text{ (K)} = t \text{ (}^\circ\text{C)} + 273.15$. Differences are the same in K and $^\circ\text{C}$; ratios are not.

Three kinds of claim

Claimed η compared with η_{Carnot}	Verdict
$\eta_{\text{claimed}} > 1 - T_2/T_1$	Impossible – violates the second law
$\eta_{\text{claimed}} = 1 - T_2/T_1$	Reversible engine – ideal limit, not attainable in practice
$\eta_{\text{claimed}} < 1 - T_2/T_1$	Possible – an irreversible (real) engine

Summary

- Carnot theorem: $\eta_{\text{irrev}} \leq \eta_{\text{rev}}$ between the same reservoirs – proved by building a PMM2
- All reversible engines between the same reservoirs: same efficiency, independent of substance
- $\eta_{\text{rev}} = f(T_1, T_2) \rightarrow$ Kelvin scale:
 $(Q_1/Q_2)_{\text{rev}} = T_1/T_2$
- Triple point of water = 273.16 K fixes the unit
- $\eta_{\text{rev}} = 1 - T_2/T_1$; $(\text{COP})_{\text{ref}} = T_2/(T_1 - T_2)$; $(\text{COP})_{\text{HP}} = T_1/(T_1 - T_2)$
- Use: feasibility tests, least heat rejection, least power – always in kelvin

Next: The Clausius inequality and entropy

- $\oint \delta Q/T \leq 0$ for any cycle
- Entropy as a property; the T–s diagram
- Numerical problems: Worked Examples 2.1 – 2.6 in the lecture notes